AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M348
Applied Statistical Modelling

Arab Open University
Arab Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14/Apr./2026

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: 30
- This exam consists of **three** parts
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed

CamScanner

PART 1: MCQ Questions [5x1=5 marks]

For each of the following multiple-choice questions, select the most appropriate answer. Only one answer is correct for each question.

1) Which statement correctly describes backward selection?

   A. Starts with no variables and adds predictors
   B. Removes variables randomly
   C. Begins with all predictors and removes the least significant ones
   D. Always produces the best model

2) What is the main issue caused by multicollinearity?

   A. It improves model interpretability
   B. It makes coefficient estimates unstable
   C. It reduces training error significantly
   D. It eliminates bias

3) In simple linear regression, what does the coefficient $\beta 1$ represent?

   A. The intercept of the model
   B. The variance of the response variable
   C. The change in Y for a one-unit change in X
   D. The total variation explained by the model

4) Which of the following is a sign of overfitting?

   A. High training error and high test error
   B. Low training error and low test error
   C. High training error and low test error
   D. Low training error and high test error

5) Why might a linear regression model be preferred over a highly flexible model?

   A. It guarantees the lowest training error
   B. It avoids all assumptions about the data
   C. It is more interpretable and easier to explain
   D. It always captures nonlinear relationships

PART 2: Multiple Linear Regression for the Olympic dataset [13 marks]

ALL QUESTIONS ARE REQUIRED

The Olympic dataset provides information on nations and their performance in the Summer Olympics for the years 1996, 2000, 2004, 2008, 2012, and 2016. It includes key details such as medal counts, hosting status, and economic indicators. Specifically, the dataset includes the following variables:

- Country: The name of the country.

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

- Country Code: The three-letter ISO 3166-1 alpha-3 code representing the country.
- Year: The Olympic year in which the data is recorded.
- Medals: The total number of medals won by the country in that year.
- LagMedals: The number of medals the country won in the previous Olympic Games.
- Host : Indicates whether the country hosted the Olympic Games in the given year (1 = Yes, 0 = No).
- NextHost : Indicates whether the country will host the next Olympic Games (1 = Yes, 0 = No).
- Population: The estimated population of the country (in millions).
- Gdp: The Gross Domestic Product (GDP) of the country (in trillions of USD).

You are asked to fit a multiple linear regression model where the response variable is medals and the explanatory variables are lagMedals, population, gdp, logPopulation, logGdp, host, nextHost and year.

```
Residuals:
    Min      1Q  Median      3Q     Max
-24.9484  -0.5646  -0.0835  0.3376 29.1619

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  27.3615418 28.6230192   0.956  0.33933
lagMedals     0.9318445  0.0090753 102.679  < 2e-16 ***
population    0.0015972  0.0008645   1.848  0.06494 .
gdp           0.0038002  0.0073374   0.518  0.60463
logPopulation 0.1498995  0.0538093   2.786  0.00544 **
logGdp        0.1610786  0.1046385   1.539  0.12401
host         12.8252200  1.3109562   9.783  < 2e-16 ***
nextHost      8.6003062  1.2876444   6.679 3.87e-11 ***
year         -0.0137590  0.0142696  -0.964  0.33516
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.091 on 1060 degrees of freedom
Multiple R-squared:  0.9476,    Adjusted R-squared:  0.9472
F-statistic:  2396 on 8 and 1060 DF,  p-value: < 2.2e-16
```

1. Fit the full model, storing the results in an object called `medalsFullModel`. Display a summary for the results from fitting this model.
2. From the above summary, does it appear that there might be a more parsimonious model than the full model? Why or why not?
3. Using the results obtained above, what is the fitted equation for this model?
4. Fit the null model for the response variable `medals`, storing the results in an object called `medalsNullModel`. Display a summary of the results from fitting the model.

Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

5. Interpret the coefficient of host in the model. What does this value mean in the context of Olympic competition?

6. Write the code to run both forward and backward stepwise regression, starting from the full model to find a parsimonious model that includes some, or all, of the variables: `lagMedals`, `population`, `gdp`, `logPopulation`, `logGdp`, `host`, `nextHost` and `year`
7. Based on the stepwise regression results, which variables are included in the final model?

	Df	Sum of Sq	RSS	AIC
<none>			10141	2419.1
+ year	1	9	10132	2420.1
- population	1	32	10173	2420.5
+ gdp	1	3	10138	2420.8
- logPopulation	1	73	10214	2424.8
- logGdp	1	74	10215	2424.9
- nextHost	1	428	10569	2461.3
- host	1	916	11057	2509.5
- lagMedals	1	100861	111002	4975.2

8. Write the code to run both forward and backward stepwise regression, starting from the null model.
9. What does the adjusted R^2 tell us about the model, and why is adjusted R^2 more reliable than multiple R^2 when comparing models?
10. The residual standard error (RSE) is equal to 3.091. Suppose a previous multiple regression model with all the explanatory variables except the host variable had a much higher RSE. What does this reduction suggest about the model's improvement?
11. What are the two types of transformations that can be applied to enhance the normality assumptions?
12. If the distribution of the variable medals shows a long right tail, which type of ladder-of-powers transformation would be most appropriate, and why?
13. If the transformed model shows lower residual standard error and residuals closer to normality, what do you conclude?

PART 3: Simple Regression with a factor (FIFA dataset) [12 marks]

ALL QUESTIONS ARE REQUIRED

Fédération Internationale de Football Association, more commonly known as FIFA, describes itself as the international governing body of association football (also known as soccer). FIFA maintains a huge database containing various attributes of football players. The FIFA 19 database includes data for 2019 on more than 18 000 footballers. In this part, we will fit a regression model for the response variable **strength** from the FIFA 19 dataset, using **skillMoves** as a factor. These variables are described as:

- **strength:** a score of strength, expressed as an integer between 0 and 100
- **skillMoves:** an assessment of each player's football 'skill' moves, taking possible values 1, 2, 3, 4 and 5 (with 5 being the highest level).

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

The resulting output of the regression are given below:

```
Residuals:
    Min      1Q  Median      3Q     Max
-13.475  -3.839   0.525   3.525  11.161

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   71.250      1.563  45.587  <2e-16 ***
skillMoves2    4.607      2.130   2.163   0.033 *
skillMoves3    0.225      1.782   0.126   0.900
skillMoves4   -2.411      1.841  -1.310   0.193
skillMoves5   -5.583      3.495  -1.598   0.113
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.414 on 95 degrees of freedom
Multiple R-squared:  0.1698,    Adjusted R-squared:  0.1349
F-statistic: 4.859 on 4 and 95 DF,  p-value: 0.001312
```

1. Write the code to fit a regression model for the response variable strength using the factor skillMoves as an explanatory variable, storing the results in an object called strength1FactorFit. Obtain a summary of the results from fitting the model.
2. Interpret the baseline mean (intercept) for `strength`. What does this represent for the footballers with the lowest skill level (`skillMoves` = 1)?
3. Interpret the regression coefficient for `skillMoves2`. How does the estimated value for `skillMoves2` compare to `skillMoves` = 1? Is there a statistically significant difference?
4. Based on the given p-values and R^2, can we confidently say that skill level has a "strong" relationship with `strength`?
5. Write the regression model equation using the estimated coefficients from the output. Express the model as a single equation with `strength` as the response variable and `skillMoves` as the explanatory factor.
6. What would be the fitted value for players with `skillMoves` equal to 3?
7. Does the `skillMoves` variable appear to have a significant impact on `strength` based on the results from the `summary`? How does the p-value from the F-test for the overall model (given by `F-statistic` and `p-value`) inform you about the significance of the relationship between `strength` and `skillMoves`?
8. In the model `strength ~ skillMoves`, identify the dependent and independent variables. How are these variables structured in terms of prediction?
9. Why is it important to examine residual plots (such as checking for normality and constant variance) after fitting this regression model? What do these residuals tell you about the model's assumptions?

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

10. What are the null and alternative hypotheses for an overall F-test to assess the relationship between `strength` and `skillMoves`?
11. If the preferredFoot is: the foot preferred by each player, taking possible values left and right.
    a) Write the R code to produce the following ANOVA table for the model with the response variable **strength** and the factor **preferredFoot**.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
preferredFoot	1	4	4.17	0.122	0.728
Residuals	98	3350	34.19		

    b) The p-value associated with the F-value is reported to be very large. What do you conclude?
    c) Write the null and alternative hypotheses for the overall ANOVA F-test of this model.

End of questions

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner